

Agent-Based Infrastructures for Diachronic Digital Humanities

Cohort

Agents for retrieving, comparing and documenting textual
evidence

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Data and thanks

Thank you, Michael Radich

For sharing a CBETA-derived research corpus, working catalogue and curated vocabulary list.

368 texts/chapters in 13 vocabulary comparison groups.

The source texts derive from CBETA. This is a selected corpus, not the whole canon.

Functions

Corpus search

Passage and vocabulary comparisons

Agent investigations, source records and researcher decisions

Example research questions

Lotus Sūtra: where do the two translations discuss the same material in different words?

T0603 and its commentary: how much does repeated wording affect the vocabulary result?

These examples test the tools on known textual relationships.

Cohort

Corpus → related passages → inspect shared wording

Vocabulary comparison → repeat with a source excluded

Inquiry → saved actions → Findings → Graph

Results

Lotus Sūtra: 283 of 327 indexed passages retrieve the other translation first.

Vocabulary: Dharmarakṣa ranks first for the earlier Lotus translation under both lists.

T0603: removing the commentary changes the first-ranked group.

Limits

A similar passage still needs to be read in context.

Vocabulary rankings can be wrong. Dates and borrowing need separate evidence.

The researcher decides whether to accept a proposal.

Corpus

COHORT

beta

Graph

Findings

Corpus

Vocabulary comparison

Inquiry

Corpus

Exact phrase

Related passages

Find passages with similar content in other works. Start from an indexed text, not a typed research question.

Embedding model (from filename): mitra-qwen35-embedder
2,160 indexed texts/chapters • 54,616 passages • 400-character windows

► Model and search method

Indexed text

Source position

Lotus Sūtra 正法華經 • Dharmarakṣa • catalogue remainder • T0263

4000

Find related passages

Lotus Sūtra • Dharmarakṣa

Body, mind and senses • T0603

Maitreya • T0453

SELECTED PASSAGE • LOTUS SŪTRA 正法華經 • DHARMARAKṢA • CATALOGUE REMAINDER • T0263-REST

Source characters 4000–4400 (zero-based, end excluded)

[Source passage hidden in this screenshot]

Previous passage

Next passage

RELATED PASSAGES

Showing 5 of 54,278 candidate passages, ranked by cosine similarity.

1. Lotus Sūtra 妙法蓮華經 • Kumārajīva • without Devadatta chapter • T0262-exDevadatta

Cosine 0.865

T0262-exDevadatta • Source characters 2800–3200

[Related passage hidden in this screenshot]

► Check shared wording

2. T0278-ex-earlier-parallels

Cosine 0.856

T0278-ex-earlier-parallels • Source characters 208000–208400

[Related passage hidden in this screenshot]

Interface reference • source passages hidden where applicable

Vocabulary comparison

COHORT beta

GraphFindingsCorpus**Vocabulary comparison**Inquiry

?

⚙

Analyze this view

Compare a text's wording

Compare short-string counts across corpus groups.

Try a text: Lotus Sūtra • Dharmarakṣa Yin chi ru jing 陰持入經 • T0603 Maitreya's descent 彌勒下生經 • T0453

Examples start with the curated strings and no additional exclusions.

► Change the selected text

► Vocabulary and method

Yin chi ru jing 陰持入經 • T0603

Catalogue classification: **An Shigao 安世高 (ASg)**

9,254 Han characters (12,012 code points) • 45 matched string occurrences • 35 different strings • vocabulary: Radich's curated strings (7,270 strings) • **1 unit withheld** from the profiles

Highest-ranked comparison group	Next comparison group	Score difference
Other material before Dharmarakṣa 竺法護 (pre-Dhr- other)	An Shigao 安世高 (ASg)	+1.696
#11 of 21 in Radich's order	#1 of 21 in Radich's order	log-odds per distinct string; not comparable across texts

Inquiry

COHORT

beta

Graph

Findings

Corpus

Vocabulary comparison

Inquiry

?

⚙

Inquiry

Choose an example or write a question to begin.

1. Choose or record a research question

Cancel

Choose an example, then edit and record it.

Find related passages: T0603

T0453: find a parallel

Test recurring formulae

Research question

Which passages are related to the Yin chi ru jing (T0603), and what might explain the relationship?

Research instructions

Find related passages, compare shared wording, and consider quotation, commentary and recurring formulas. Report source positions and limitations.

Specify the approach, evidence to return and limits. These instructions are saved and sent to the agents.

Record question

► Saved questions (4)

Show hidden questions (3)

2. Start a new agent run

Findings

COHORT beta

Graph **Findings** Corpus Vocabulary comparison Inquiry

Hypotheses 8

Claims and conjectures, newest first. Not ranked by confidence.

CONJECTURE

The close relationship between Yin chi ru jing (T0603) and T1694 reflects textual dependence or shared formulaic stock — quotation, commentary, or reuse of the same base exposition — rather than independent composition; whether it also reflects a shared hand cannot be settled from this corpus alone.

ATTESTED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT PROSPECTIVE QUERY NOT YET RUN DOSSIER

CLAIM

Yin chi ru jing (T0603) and T1694 reproduce the same extended wording, including the eighteen-base exposition and long doctrinal runs up to 279 characters, with 64.45% of T0603's ten-character strings found in T1694.

ATTESTED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT

CONJECTURE

The apparent similarity between T0603 and Other material before Dharmarakṣa 竺法護 (pre-Dhr-other) translator vocabulary is driven by repeated material shared with T1694: once T1694 is withheld from the comparison profiles, the leaning for T0603 collapses to no meaningful distinction between An Shigao 安世高 (ASg) and Other material before Dharmarakṣa 竺法護 (pre-Dhr-other).

ATTESTED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT PROSPECTIVE QUERY NOT YET RUN DOSSIER

CLAIM

Yin chi ru jing (T0603) reproduces wording also found in T1694 at length, including a 128-character run enumerating the eighteen bases and a 279-character run, with 64.45% of T0603's ten-character strings found in T1694.

ATTESTED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT

CLAIM

T0603 (陰持入經) and T1694 reproduce the same extended wording verbatim, including a 128-character run beginning 四耳五聲六識 and a 279-character run on non-extinction/purity; 64.5% of T0603's ten-character strings occur in T1694.

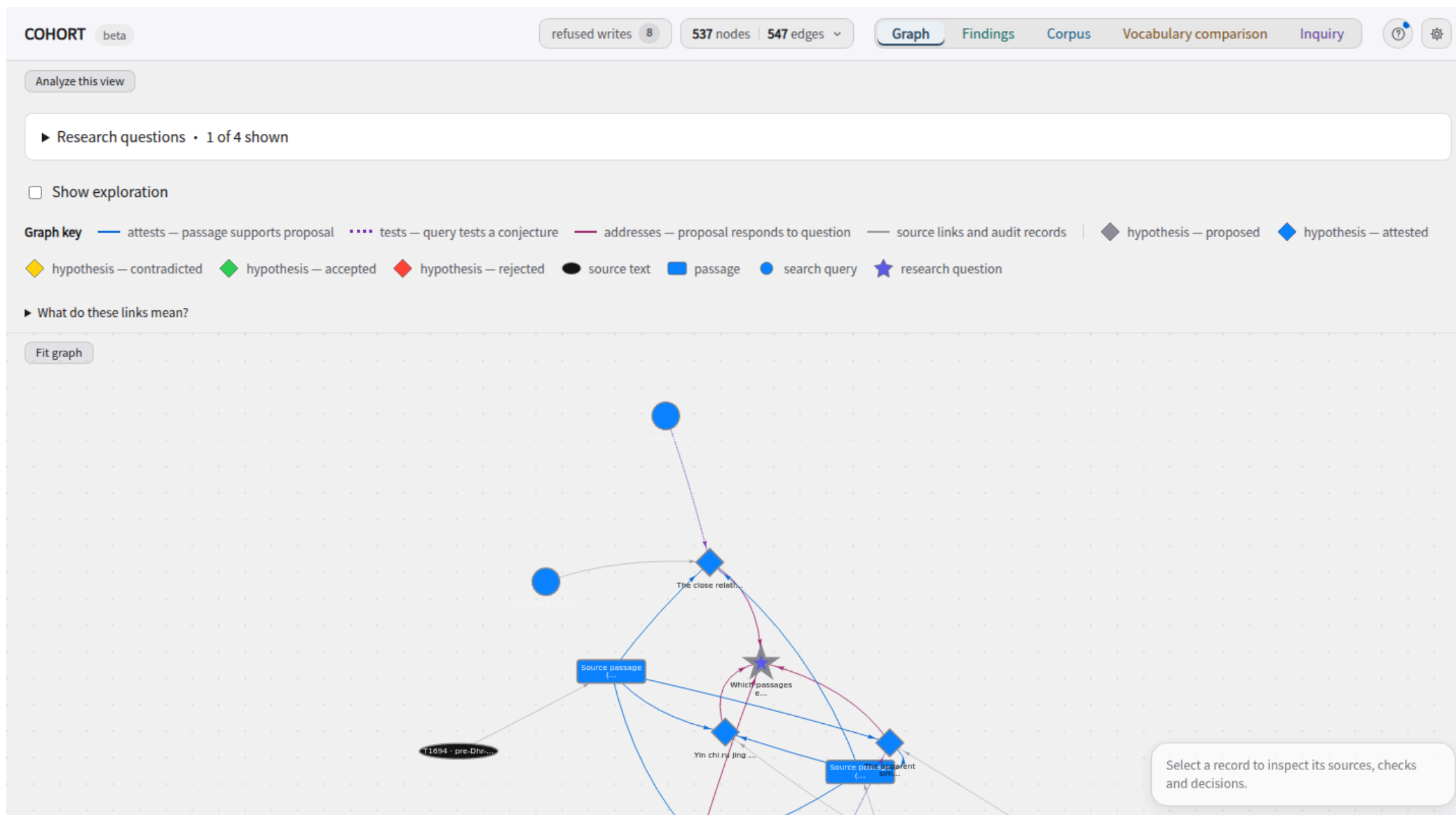
ATTESTED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT

CONJECTURE

The Other material before Dharmarakṣa 竺法護 (pre-Dhr-other) leaning of T0603 (陰持入經) under Radich's vocabulary depends on T1694 sharing its wording: with T1694 withheld the leaning disappears (An Shigao 安世高 (ASg) ahead by margin 0.096, gap 3.4, i.e. no evidence either way), so the corpus as configured cannot settle whether An Shigao translated T0603.

ATTESTED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT PROSPECTIVE QUERY NOT YET RUN DOSSIER

Graph



Translator usage or later reuse?

Which texts share wording with T0603, and does that similarity reflect a translator's usage or later reuse?

T0603 has an accepted attribution to An Shigao.
Its later commentary repeats wording from the earlier text.

Use this known case to examine the method before applying it to disputed texts.

Find a phrase and open its source

Cohort Graph Findings Corpus Evidence Inquiry

Corpus

Suggested searches

Opening formula · 如是我聞

Open a matching source record

Interface diagram

Exact wording

Read the context

Prepare an inquiry

Give the agent a question and instructions

Cohort	Graph	Findings	Corpus	Evidence	Inquiry
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Inquiry

Research question

Does T0603's shared wording reflect a translator's usage or later reuse?

Research instructions

Find passages; compare reuse and translator usage.

Record question

Inquire

Question: what to investigate

Instructions: how to approach it

Run results: what the agents did

Interface diagram

Retrieve candidates by content

Input	A text or passage
Semantic neighbours	Retrieve similar passages from the embedding index
Researcher's next step	Read candidates and choose pairs to compare

Similar content can come from a shared subject, a formula or textual reuse.

Locate shared wording in a selected pair

T0603

Yin chi ru jing
陰持入經



T1694

Yin chi ru jing zhu
陰持入經註

Earlier translation → later commentary: a relationship known from scholarship.

Alignment locates shared wording; it does not establish the chronology.

Compare string counts across groups

Cohort	Graph	Findings	Corpus	Evidence	Inquiry
Vocabulary comparison					
Text: T0603 · 陰持入經					
Vocabulary and method					
Radich's curated strings					
All comparison groups, ranked					
Rank	Group	Score relative to leader			

Select the strings

Read the ranked groups

Inspect the wording
behind the scores

Interface diagram

Repeat T0603 without its commentary

T1694 belongs to “Other material before Dharmarakṣa” in this catalogue.

With T1694

Other material before
Dharmarakṣa 竺法護

Lead over next group: 1.696

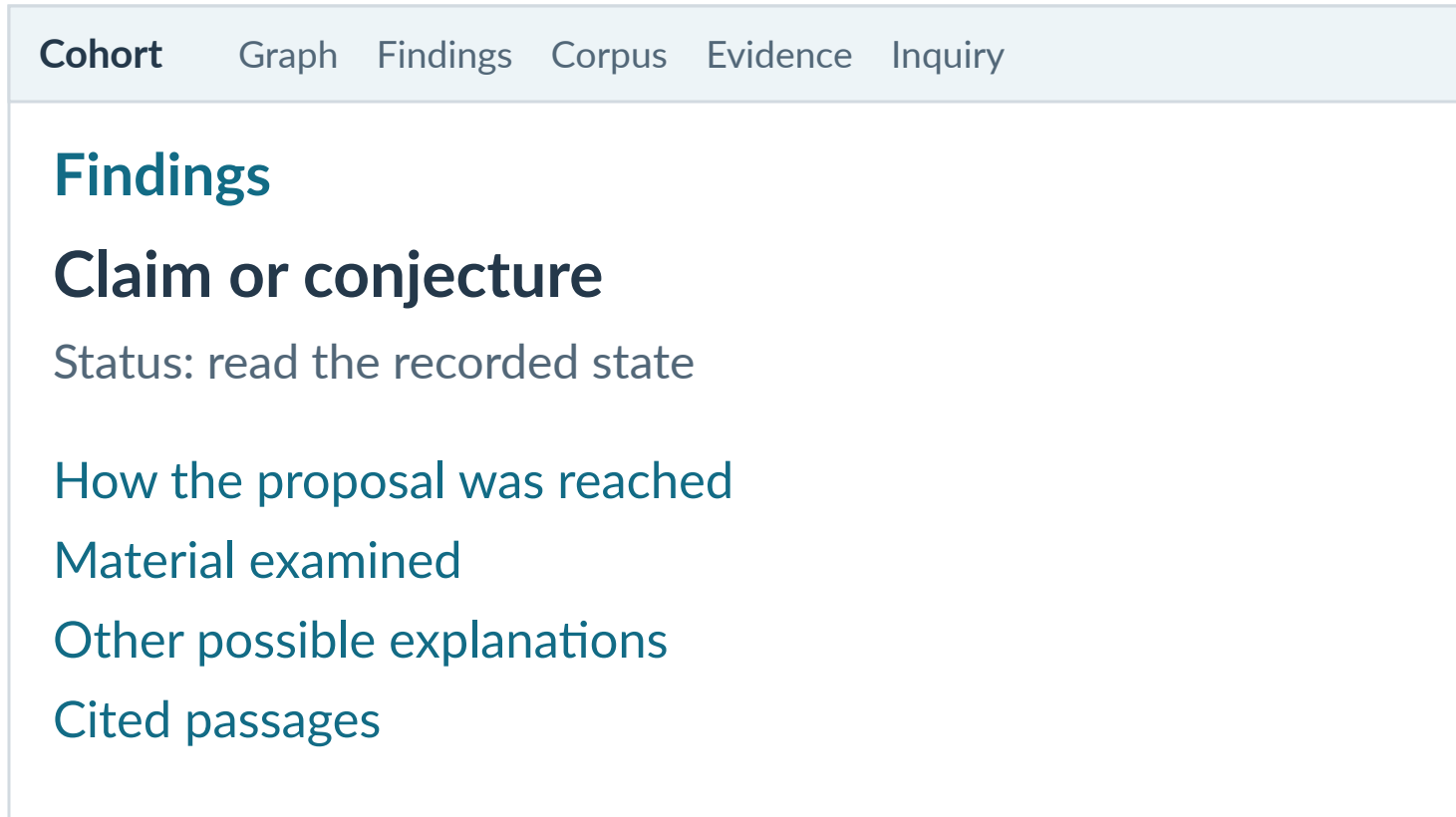
Without T1694

An Shigao 安世高

Lead over next group: 0.096

Removing the later commentary brings the prediction into agreement with the accepted attribution.

Read the proposal and its alternatives



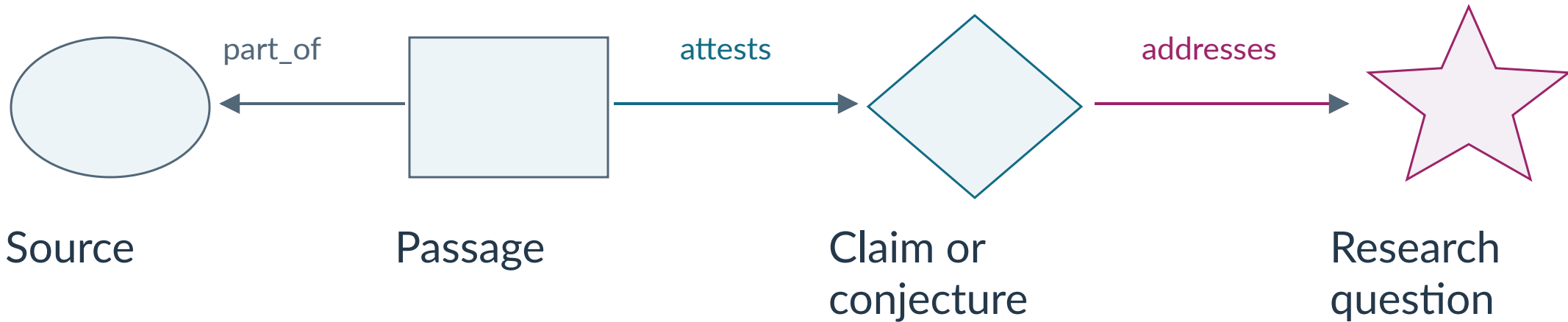
Interface diagram

Expand one proposal

Compare its explanation
with the sources

Follow a citation into
Graph

Follow a proposal back to its sources



Select a record to inspect its worker, model, citations and checks.

Separate citation checks from researcher decisions

Span verification

Was the quotation found in the source?

Review

What did a separate agent check and report?

Researcher decision

Should this particular record be accepted or rejected?

Accepting a passage does not accept every claim connected to it.

Ask for an explanation of the current view

Analyze this view

Open the side panel; choose a model

Start analysis

Inspect records, retrieve passages and repeat comparisons

Saved analyses

Reopen the explanation and its actions

The analyst cannot modify graph records. Its explanation refers to the selected snapshot.

Historical information and measured results

From scholarship

Attributions, approximate dates and the commentary relationship

From Cohort

Located passages, shared wording and the effect of excluding a source

Further research

Whether the wording supports an ascription, revision or reuse hypothesis

Research tasks to develop with scholars

Find useful parallels

Which retrieved passages deserve close reading?

Test a comparison

Does it persist across genre, length and source exclusions?

Assess an ascription

Which observations could distinguish competing explanations?

Choosing a vocabulary

Curated strings

Built for work on a Dharmarakṣa dictionary; uneven coverage

Frequent corpus strings

Selected without translator labels; can favour genre formulas

Combined lists

Union of both lists; their biases remain

The comparison groups

An Shigao 安世高

Baoyun 寶雲

Buddhabhadra 佛陀跋陀羅

Dharmakṣema 曇無讖

Dharmarakṣa 竺法護

Guṇabhadra 求那跋陀羅

Kumārajīva 鳩摩羅什

Lokakṣema 支婁迦讖

Gautama Saṅghadeva 僧伽提婆

Zhu Fonian 竺佛念

Zhi Qian 支謙

Mixed categories

Other material before Dharmarakṣa 竺法護

Material after Lokakṣema 支婁迦讖

Two score displays, two scales

Full ranking

Group score – highest score

Leader = 0; others are negative

Headline margin

Highest score – next score

Distinct matched strings

Neither display gives a probability that a translator is correct.

Another retrieval and alignment example

T0453 • Maitreya's descent
彌勒下生經

Known parallel in the Ekottarikāgama
增壹阿含經 • T0125, section 48.3

Use the located parallel to inspect what retrieval and alignment return.

Q3: disputed Paramārtha ascriptions

Do disputed texts resemble Paramārtha's accepted translations or other textual traditions?

Compare candidate features against benchmark and control works.
Inspect whether reuse, genre or revision explains the resemblance.

Separate development branches; results and historical explanations still need review.

The names used in the T0603 comparison

An Shigao 安世高

Second century CE; T0603 is traditionally associated with him

Dharmarakṣa 竺法護

Late third / early fourth centuries CE; a later reference point

T1694 • 陰持入經註

Commentary on T0603; assigned to the mixed pre-Dharmarakṣa group

The research activities and their controls

Corpus retrieval

Corpus search and agent retrieval tools

Semantic analysis

Semantic neighbours and wording comparisons

Alignment

Shared passages with source positions

Interpretation

Proposals in Findings; read-only AI analysis

Provenance and decisions

Graph records, checks and researcher actions

How each tab works

Corpus	Exact search, stored-vector retrieval and shared wording.
Vocabulary comparison	String counts scored against catalogue-labelled groups.
Inquiry	A worker calls tools; a separate reviewer checks proposals.
Findings	Read saved proposals, citations and checks.
Graph	Inspect links and record researcher decisions.

Which model does what?

Embedder	mitra-qwen35-embedder (identified by the index filename).
Passage retrieval	Cosine similarity between stored passage vectors.
Worker	Chooses tools and writes proposals. See Inquiry for its model.
Reviewer	A different model family records checks on proposals.
No embedding call	Exact search, vocabulary rankings and graph display.

Reading the Graph arrows

Witness → Passage

contains: the passage belongs to this source.

Passage → Proposal

attests: the passage is offered as support.

Query → Proposal

searched for: a search used to prepare the proposal.

Query → Conjecture

tests: a planned check; inspect whether it was run.

Proposal → Question

addresses: the question this proposal concerns.